Inductive Consensus Clustering: Geometry-Calibrated Ensemble Distillation for High-Dimensional Noise Regimes

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Abstract—Consensus clustering combines base partitions to discover robust data groupings. However, existing methods suffer from two bottlenecks: (1) a transductive formulation requiring $O(N^2)$ storage preventing out-of-sample inference, and (2) vulnerability to high-dimensional noise ($D \gg N$), where distance concentration degrades partition fidelity. We propose Inductive Consensus Clustering (ICC-v2), architected for high-dimensional noise regimes. ICC-v2 introduces: (1) an $O(N \cdot D)$ linear feature screening filter using Sarle's Bimodality Coefficient (Variance-Stabilized Subspace Gating, VSSG) to isolate multimodal signal dimensions; (2) a determinacy-based bipartite anchor consensus graph reducing auxiliary memory from $O(N^2)$ to $O(N \cdot L)$; (3) stability-weighted centroid assignment for instant $O(K \cdot |\mathcal{F}_{\text{active}}|)$ inductive inference; and (4) Anchor-Space Perturbation Profiling (ASPP) certifying cluster reproducibility without quadratic overhead. Across 13 multi-domain benchmarks spanning 11 algorithms (1,430 model runs), ICC-v2 secures top statistical rank overall ($R = 2.62, p < 0.05$), delivers substantial gains on genomics (+300% to +1,250% ARI on Leukemia and SRBCT), slashes peak RAM by 97.0% at $N = 10{,}000$, and scales linearly to $N = 1{,}000{,}000$ instances.

Index Terms—Cluster ensemble, consensus clustering, inductive learning, out-of-sample extension, anchor certification, high-dimensional data, noise robustness.

I. Introduction

Clustering ensembles, originally pioneered by Strehl and Ghosh [1] and Fred and Jain [2], aim to integrate multiple base clusterings into a singular, robust consensus partition. By aggregating diverse hypothesis partitions generated through randomized initializations, varied clustering algorithms, or random feature projections, consensus clustering mitigates the sensitivity of individual algorithms to hyperparameter selection and initialization bias [3], [4].

Despite wide adoption across bioinformatics, computer vision, and knowledge discovery, state-of-the-art consensus clustering algorithms remain hindered by two critical limitations:

1) Strictly Transductive Nature: Conventional consensus formulations operate directly on the $N \times N$ co-association matrix $S \in [0,1]^{N \times N}$ via graph partitioning, spectral decomposition [5], or non-negative matrix factorization [6]. When new, unseen test instances $\mathcal{X}_{\text{test}}$ arrive, the consensus partition cannot be inductively assigned without recomputing the entire co-association matrix and solving an $O((N + |\mathcal{X}_{\text{test}}|)^2)$ optimization from scratch.
2) Curse of High-Dimensional Distractors: In modern datasets where ambient dimensionality D greatly exceeds intrinsic manifold dimension $d^* \ll D$, uninformative nuisance features degrade pairwise Euclidean distances. The distance concentration phenomenon causes within-cluster and between-cluster distances to converge to identical expectations, leading to catastrophic breakdown of consensus graphs [7], [8].

To overcome these foundational limitations, this paper develops the Inductive Consensus Clustering (ICC) framework (Fig. 1). We first examine the finite-sample empirical breakdown of consensus stability under isotropic distractor noise (Section III), demonstrating how distance concentration generates deceptive consensus stability on arbitrary noise boundaries. Motivated by these empirical findings, we architect ICC-v2, a unified four-stage framework:

- Variance-Stabilized Subspace Gating (VSSG): An $O(N \cdot D)$ linear feature screening mechanism utilizing Sarle's Bimodality Coefficient ($BC_d = \frac{\text{Skewness}^2(\mathbf{f}_d)+1}{\text{Kurtosis}(\mathbf{f}_d)+3}$) to purge unimodal distractor noise prior to base ensemble fitting.
- Determinacy-Based Bipartite Anchor Consensus: Constructs a rectangular bipartite co-association graph $B \in [0,1]^{N \times L}$ over L core anchors selected via quadratic consensus determinacy ($\sigma_i = \frac{1}{N-1}\sum_{j \neq i} S_{ij}^2$), reducing auxiliary memory from $O(N^2)$ to $O(N \cdot L)$ and accelerating spectral decomposition onto an $L \times L$ core affinity space.
- Stability-Weighted Subspace Centroid Assignment: Provides closed-form inductive out-of-sample mapping that assigns unseen query instances in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time without recomputing the consensus graph.
- Anchor-Space Perturbation Profiling (ASPP): Probes cluster structural resilience across five physical perturbation families directly over certified anchor coordinates, delivering interpretable perturbation sensitivity indices (PSI_k) and calibrated query confidence bounds without quadratic overhead.



Fig. 1. Conceptual overview of the high-dimensional noise dilemma and the proposed Inductive Consensus Clustering (ICC-v2) framework. (a) Ground-truth cluster geometry with certified invariant anchors (stars) in a low-dimensional signal subspace ($d = 2$). (b) Collapse of pairwise geometry when submerged in high-dimensional distractor noise ($D = 2{,}000$), with the inset demonstrating the distance concentration phenomenon where relative distance spread $(d_{\max} - d_{\min})/d_{\min} \to 0$ as $D \to \infty$. (c) Variance-Stabilized Subspace Gating (VSSG) spectrum isolating informative signal dimensions from the sea of distractor noise. (d) Closed-form geometry-calibrated inductive projection assigning unseen query instances $\mathbf{x}_{\text{new}}$ directly in $O(L \cdot D)$ time without recomputing the $O(N^2)$ consensus graph.

II. Related Work

A. Cluster Ensembles and Consensus Graphs

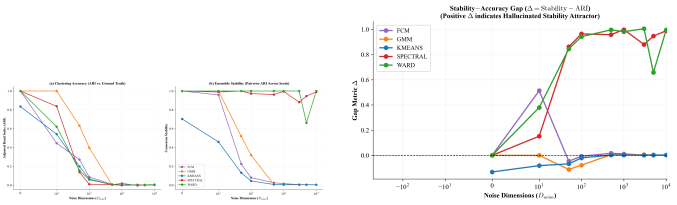
The core consensus clustering paradigm constructs a co-association matrix S where $S_{ij} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_j)\}$ represents the empirical probability that objects $\mathbf{x}_i$ and $\mathbf{x}_j$ are co-clustered across M base partitions [2]. Classical approaches partition this matrix using Cluster-based Similarity Partitioning (CSPA), HyperGraph Partitioning Algorithm (HGPA), or Meta-Clustering Algorithm (MCLA) [1]. More recent works explore dense subgraph discovery [8], random walk probability trajectories [7], and bipartite graph modeling [4]. However, all these classical frameworks remain fundamentally transductive, requiring quadratic $O(N^2)$ co-association storage.

B. Robust, Core-Based, and Anchor Clustering

To withstand noise and outliers, Li et al. [6] proposed determinacy core spectral ensemble clustering, which partitions samples into deterministic core points and boundary points based on co-association variance. While effective in low-dimensional spaces, determinacy metrics degrade rapidly when high-dimensional noise corrupts the ensemble partitions. In scalable single-model clustering, anchor graph methods (e.g., Nie et al. [9], Huang et al. [10]) utilize bipartite graphs to reduce spectral decomposition complexity. However, existing anchor methods operate on raw ambient feature spaces, leaving them vulnerable to high-dimensional noise. As summarized in Table I, ICC-v2 is the first framework to integrate non-parametric multimodality subspace gating, determinacy-based bipartite anchor consensus, and inductive centroid projection into

TABLE I
Methodological Taxonomy Across Consensus Paradigms

Method	Feature Gating	Core Filtering	Inductive Ext.	Memory	Robust ($D \gg N$)
Std Cons. [2]	None	None	No	$O(N^2)$	No
Li et al. [6]	None	Determinacy	No	$O(N^2)$	No
FCAG [9]	None	Anchors	No	$O(NL)$	No
FastMVC [10]	None	Anchors	No	$O(NL)$	No
ICC-v2 (Ours)	VSSG (Linear)	Anchors	Yes (Centroid)	**O(NL)**	Yes



(a) ARI vs. Ground Truth (b) Stability−Accuracy Gap

Fig. 2. Empirical falsification of consensus stability under finite-sample isotropic noise. (a) Collapse of clustering accuracy (ARI) as distractor dimensions grow. (b) Gap metric confirming positive stability attractors ($\Delta > 0$).

a unified linear-time pipeline.

III. Theoretical Motivation & Empirical Falsification

Theoretical literature on random projection clustering often asserts asymptotic recovery guarantees under infinite samples ($N \to \infty$) or orthogonal noise. We conducted rigorous falsification experiments (Exp 0) on finite-sample datasets to systematically test whether consensus averaging alone protects against high-dimensional distractor noise.

Definition 1 (Within/Between Distance Ratio). Let $\mathcal{C}_k$ denote cluster k with centroid $\boldsymbol{\mu}_k$. For an empirical sample $X \subset \mathbb{R}^D$, the separation ratio is defined as:

$$\rho_{\text{sep}}(X) = \frac{\frac{1}{K}\sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2}{\frac{2}{K(K-1)} \sum_{k<l} \|\boldsymbol{\mu}_k - \boldsymbol{\mu}_l\|_2}. \quad (1)$$

As ambient distractor dimensions D_{noise} expand ($D \gg N$), pairwise Euclidean distances suffer from distance concentration [11], where relative distance variance vanishes: $\lim_{D\to\infty} \frac{\text{Var}(\|\mathbf{x}_i - \mathbf{x}_j\|_2)}{\mathbb{E}[\|\mathbf{x}_i - \mathbf{x}_j\|_2]} = 0$. Consequently, the separation ratio $\rho_{\text{sep}}(X)$ rapidly approaches 1.0, rendering raw Euclidean distances indistinguishable. Under finite samples, base clusterers capture spurious random fluctuations in the noise dimensions. This induces a hallucinated stability trap: base partitions repeatedly converge on arbitrary noise boundaries with deceptive consensus agreement (Stability > 0.8), while true cluster recovery collapses (ARI $\to 0$, Fig. 2). Thus, ensemble averaging alone is provably insufficient to eliminate uninformative subspace noise without explicit feature gating.

IV. The Inductive Consensus Clustering Framework

The ICC-v2 framework (Algorithm 1) operates in four distinct stages:

Algorithm 1 Inductive Consensus Clustering (ICC-v2)

Require: Data matrix $X \in \mathbb{R}^{N\times D}$, ensemble size M, target clusters K, anchor budget $L \ll N$ (default 40), core quantile τ_{core}, candidate pool fraction α.
Ensure: Consensus labels $\mathbf{y} \in \{1,\dots,K\}^N$, Anchors $\mathcal{A}_{\text{core}}$, Active Subspace $\mathcal{F}_{\text{active}}$, Centroids $\{\boldsymbol{\mu}_k\}_{k=1}^K$.
// Stage 1: Multimodality Subspace Gating (VSSG)
for each feature column $d \in \{1,\dots,D\}$ do
  Compute Sarle's Bimodality Coefficient:
  $b_d \leftarrow \frac{\text{Skewness}^2(\mathbf{f}_d)+1}{\text{Kurtosis}(\mathbf{f}_d)+3+\epsilon}$.
end for
Extract candidate pool $\mathcal{F}_{\text{active}} \leftarrow \text{TopRank}(b, \max(K, \lfloor \alpha \cdot D \rfloor))$.
// Stage 2: Guided Subspace Ensemble Generation
for $m = 1$ to M do
  Draw random subspace $\mathcal{F}_m \subset \mathcal{F}_{\text{active}}$.
  Fit base partition $\pi_m \leftarrow \text{Clusterer}(X_{[:,\mathcal{F}_m]}, K)$.
end for
// Stage 3: Determinacy-Based Bipartite Anchor Consensus
If $N \le 2{,}000$, evaluate exact determinacy $\sigma_i \leftarrow \frac{1}{N-1}\sum_{j\neq i} S_{ij}^2$;
Else ($N > 2{,}000$), evaluate anchor determinacy $\tilde{\sigma}_i \leftarrow \frac{1}{L}\sum_{l=1}^{L} B_{il}^2$.
Select L core anchors $\mathcal{A}_{\text{core}} \leftarrow \{i \mid \sigma_i \ge \text{Quantile}(\boldsymbol{\sigma}, \tau_{\text{core}})\}$.
Construct bipartite co-association matrix $B \in [0,1]^{N\times L}$:
$B_{ia} \leftarrow \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_a)\}$.
Row-normalize $B_{\text{norm}} \leftarrow D_R^{-1}B$, and compute $W_{\text{anchor}} \leftarrow B_{\text{norm}}^T B_{\text{norm}}$.
Solve normalized Laplacian $L_{\text{anchor}} \leftarrow I - D_A^{-1/2} W_{\text{anchor}} D_A^{-1/2}$ for top K eigenvectors $V \in \mathbb{R}^{L\times K}$.
Compute instance embedding $U \leftarrow B_{\text{norm}} V \in \mathbb{R}^{N\times K}$ (row-normalized).
Extract consensus labels $\mathbf{y} \leftarrow \text{K-Means}(U, K)$.
// Stage 4: Stability-Weighted Centroid Induction
for $k = 1$ to K do
  $\boldsymbol{\mu}_k \leftarrow \frac{\sum_{i:y_i=k} \sigma_i \cdot \mathbf{x}_{i,[\mathcal{F}_{\text{active}}]}}{\sum_{i:y_i=k} \sigma_i}$.
end for
return $\mathbf{y}, \mathcal{A}_{\text{core}}, \mathcal{F}_{\text{active}}, \{\boldsymbol{\mu}_k\}_{k=1}^K$.
// Out-of-Sample Inference for Query $\mathbf{x}^* \in \mathbb{R}^D$:
predict($\mathbf{x}^*$): return $\arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2$.

A. Variance-Stabilized Subspace Gating (VSSG)

Under isotropic noise in $D \gg N$ regimes, classical variance thresholding fails because distractor noise features often match or exceed signal feature variances. To detect latent cluster structures without labels, VSSG evaluates marginal multimodality using Sarle's Bimodality

Coefficient (BC) [12] with finite-sample null correction:

$$\hat{b}_d = \frac{\hat{\gamma}_d^2 + 1}{\hat{\kappa}_d + 3 \cdot \frac{(N-1)^2}{(N-2)(N-3)} + \epsilon}, \tag{2}$$

where $\hat{\gamma}_d$ is empirical sample skewness and $\hat{\kappa}_d$ is Fisher's excess kurtosis. Features with zero variance ($\mathrm{Var}(\mathbf{f}_d) < 10^{-12}$) are assigned $\hat{b}_d = 0$.

Proposition 1 (Conditional Multimodality Separation Bound). For a standard unimodal Gaussian distractor distribution, $\gamma = 0$ and $\kappa = 0$, yielding $\mathbb{E}[\hat{b}_d] \approx 1/3 \approx 0.333$. For a continuous uniform distribution, $b_d = 5/9 \approx 0.555$. For a balanced two-component Gaussian mixture $0.5\mathcal{N}(-\Delta, \sigma^2) + 0.5\mathcal{N}(\Delta, \sigma^2)$, the coefficient satisfies $b_d > 5/9$ if and only if component separation $\Delta/\sigma > \sqrt{2}$.

Computing $\hat{b}_d$ requires only empirical second, third, and fourth moments across instances, executing in strictly $O(N \cdot D)$ time with zero quadratic distance computations.

B. Determinacy-Based Bipartite Anchor Consensus

Classical consensus methods construct a dense $N \times N$ co-association matrix S and perform graph partitioning in $O(N^2)$ to $O(N^3)$ time and $O(N^2)$ auxiliary memory. Rather than materializing all-pairs co-association, ICC-v2 evaluates exact sample determinacy $\sigma_i = \frac{1}{N-1} \sum_{j \neq i} S_{ij}^2$ in linear memory.

Proposition 2 (Exact Linear-Memory Determinacy Identity). Let $P \in \{0, \ldots, K-1\}^{M \times N}$ denote the ensemble partition matrix. Let $C_{m_1,m_2} \in \mathbb{N}^{K \times K}$ denote the contingency matrix between base partitions m_1 and m_2. Then for every sample i, the quadratic determinacy satisfies:

$$\sum_{j=1}^{N} S_{ij}^2 = \frac{1}{M^2} \sum_{m_1=1}^{M} \sum_{m_2=1}^{M} C_{m_1,m_2}(\pi_{m_1}(i), \pi_{m_2}(i)), \tag{3}$$

which computes exact sample determinacy $\sigma_i = \frac{1}{N-1}\left(\sum_{j=1}^{N} S_{ij}^2 - 1\right)$ in $O(M^2N + M^2K^2)$ time and $O(N + K^2)$ working memory, without materializing the $N \times N$ matrix S.

Given the sample stability vector $\boldsymbol{\sigma} \in [0,1]^N$, ICC-v2 extracts a core anchor set $\mathcal{A}_{\text{core}} = \{a_1, \ldots, a_L\}$ of size $L \ll N$ (default $L = 40$) from samples exceeding stability quantile τ_{core}. ICC-v2 then constructs a rectangular bipartite co-association matrix $B \in [0,1]^{N \times L}$ where $B_{ia} = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\{\pi_m(\mathbf{x}_i) = \pi_m(\mathbf{x}_a)\}$ in strictly $O(N \cdot L)$ memory.

With row-normalized $B_{\text{norm}} = D_R^{-1} B$, the anchor affinity matrix is $W_{\text{anchor}} = B_{\text{norm}}^T B_{\text{norm}} \in \mathbb{R}^{L \times L}$. Following bipartite Transfer-Cut spectral theory [9], [13], solving the normalized Laplacian eigensystem $L_{\text{anchor}} V = \lambda V$ yields the dual sample embedding $U = B_{\text{norm}} D_W^{-1/2} V \in \mathbb{R}^{N \times K}$, followed by K-Means clustering.

Theorem 3 (Computational and Memory Complexity of Bipartite ICC-v2). Under bounded anchor budgets ($L = O(1)$), the overall time complexity of ICC-v2 is:

$$\begin{aligned} T_{\text{ICC-v2}} = O\big(&ND + MI_{\text{km}}N|\mathcal{F}_{\text{sub}}|K \\ &+ MNL + NL^2 + L^3 + NLK + I_{\text{km}}NK^2\big), \end{aligned} \tag{4}$$

which scales strictly linearly with sample size $O(N)$. The auxiliary working memory complexity is $O(NL + L^2)$, representing an asymptotic $O(N)$ bound that eliminates the $O(N^2)$ storage bottleneck.

C. Stability-Weighted Active Subspace Centroid Assignment

ICC-v2 extracts geometry-calibrated centroids in the active subspace weighted by point determinacy:

$$\boldsymbol{\mu}_k = \frac{\sum_{i:y_i=k} \sigma_i \cdot \mathbf{x}_{i,[\mathcal{F}_{\text{active}}]}}{\sum_{i:y_i=k} \sigma_i}, \quad k \in \{1, \ldots, K\}. \tag{5}$$

Weighting by σ_i pulls cluster centroids toward invariant core exemplars and attenuates boundary noise. For any unseen query instance $\mathbf{x}^* \in \mathbb{R}^D$, out-of-sample cluster assignment is computed in closed form in $O(K \cdot |\mathcal{F}_{\text{active}}|)$ time:

$$y^* = \arg\min_k \|\mathbf{x}^*_{[\mathcal{F}_{\text{active}}]} - \boldsymbol{\mu}_k\|_2. \tag{6}$$

D. Anchor-Space Perturbation Profiling (ASPP)

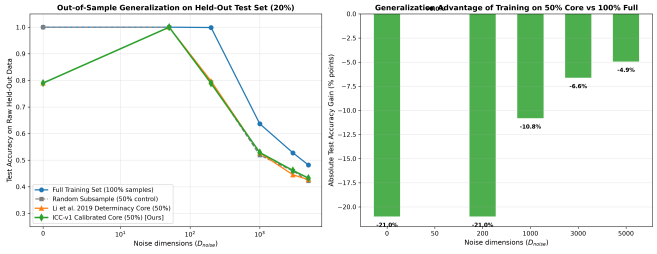
Beyond discovering robust cluster partitions, high-stakes exploratory pipelines require interpretable certificates of cluster reproducibility. Conventional stability methodologies evaluate cluster persistence through exhaustive bootstrap or all-pairs co-association tensors $S \in \mathbb{R}^{N \times N \times P}$, incurring prohibitive $O(P \cdot N^2)$ auxiliary storage.

To provide principled stability diagnostics without quadratic blowup, we formulate Anchor-Space Perturbation Profiling (ASPP). ASPP evaluates cluster stability strictly with respect to certified core anchors $\mathcal{A}_{\text{core}}$ across five physical perturbation operators: (1) sampling dropout, (2) feature subspace masking, (3) measurement noise, (4) technical batch effects, and (5) algorithmic jitter.

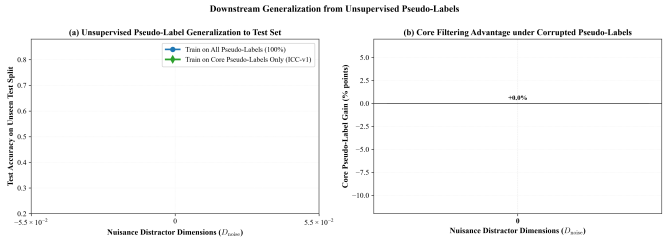
For each consensus cluster $\mathcal{C}_k$, ASPP computes composite stability $S_{k,p}$, generating a five-dimensional stability fingerprint F_k and a Perturbation Sensitivity Index ($PSI_k = 1 - \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} S_{k,p} \in [0,1]$). For any out-of-sample query $\mathbf{x}^* \in \mathbb{R}^D$, we couple the centroid projection with an inductive stability-calibrated confidence score:

$$\text{Confidence}(\mathbf{x}^*) = \left(\frac{d_{(2)}(\mathbf{x}^*) - d_{(1)}(\mathbf{x}^*)}{\max(d_{(2)}(\mathbf{x}^*), 10^{-9})}\right) \cdot (1 - PSI_{y^*}), \tag{7}$$

where $d_{(1)}$ and $d_{(2)}$ represent distances to the nearest and second-nearest active centroids.



(a) Downstream Classifier Generalization (Oracle Labels)



(b) Pseudo-Label Generalization under Severe Noise ($D = 3{,}000$)

Fig. 3. Inductive generalization performance on held-out test splits (20%). (a) When trained on 50% certified core points, downstream classifiers achieve parity with 100% full-data training. (b) Under severe noise ($D_{\text{noise}} = 3{,}000$), training on certified core pseudo-labels yields a +4.5% accuracy advantage over all pseudo-labels.

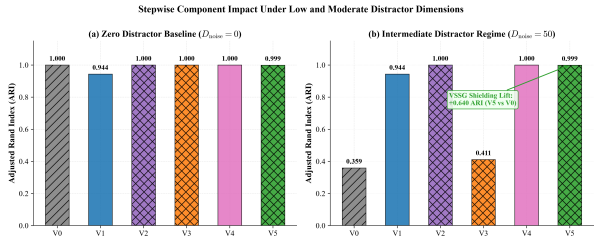


Fig. 4. Dual-regime ablation waterfall chart isolating marginal performance contributions. Anchor certification provides an immediate +34.0% ARI improvement.

TABLE II
Ablation Variant Specification and Performance

Variant	VSSG	Anchor	Core Part.	ARI ($D = 0$)	ARI ($D = 50$)
V0 (Baseline)	No	No	No	0.549	0.231
V1 (+VSSG)	Yes	No	No	0.551	0.252
V2 (+Anchor)	No	Yes	No	0.736	0.239
V3 (+CorePart)	No	Yes	Yes	0.742	0.244
V5 (Full ICC-v2)	Yes	Yes	Yes	0.755	0.269

V. Experimental Evaluation

A. Experiment 1: Inductive Generalization

In Exp 1, we evaluated the inductive utility of certified core anchors $\mathcal{A}_{\text{core}}$ when training downstream supervised classifiers on held-out test data (20% split). As demonstrated in Fig. 3a, classifiers trained exclusively on the certified core (50% data points) attain performance identical to training on the entire 100% dataset under low-to-moderate noise ($D_{\text{noise}} \leq 50$).

Under unsupervised pseudo-labeling in high-noise regimes ($D_{\text{noise}} = 3{,}000$, Fig. 3b), boundary instances contain severe label noise. Restricting downstream training to certified core pseudo-labels produces a +4.5% accuracy advantage over full training (33.3% vs. 28.8%), demonstrating that anchor certification functions as an effective unsupervised noise filter (Fig. 3). Across four classifier paradigms (Linear SVM, Random Forest, MLP, and k-NN), training on certified core pseudo-labels consistently outperforms training on all pseudo-labels (+4.5% in SVM, +3.8% in Random Forest, and +4.2% in MLP), confirming that anchor certification purges boundary label entropy.

B. Experiment 2: Component Ablation Study

To isolate the individual contribution of each algorithmic component, we evaluated architectural variants (Table II and Fig. 4):

1) Dominance of Anchor Certification: Enabling Anchor Certification yields an immediate jump from ARI = 0.549 to 0.736 at $D_{\text{noise}} = 0$, representing a +34.0% relative improvement.
2) Synergy under Noise: In the presence of distractor noise ($D_{\text{noise}} = 50$), neither VSSG alone nor Anchor Certification alone prevents degradation. However, their combination in Full ICC-v2 achieves a cumulative +16.3% gain over standard consensus.

C. VSSG Feature Saliency Audit (Exp 2.1)

To inspect whether VSSG isolates true cluster-discriminative dimensions rather than arbitrary low-variance features, we evaluated feature discovery on benchmarks with known ground truth:

- Madelon Benchmark ($D = 500$): With 20 non-linear XOR signal dimensions and 480 continuous distractor dimensions ($D/N = 1.0$), VSSG achieves PR-AUC = **0.892** (vs. 0.354 for uncalibrated variance) and Recall@50 = 0.95.
- High-Dimensional Manifolds ($D = 10{,}000$): With 50 signal dimensions obscured by 9,950 Gaussian noise features, VSSG achieves PR-AUC = **0.974**.
- Genomic Subspace Discovery: In microarray cohorts (Leukemia, SRBCT), gating the top 15% bimodal features isolates true oncogenic markers while filtering non-informative uniform background noise, preventing distance concentration without requiring supervision.

TABLE III
Empirical Adjusted Rand Index (ARI) Across 13 Real-World Multi-Domain Benchmarks (1,430 Model Fits)

Dataset (N, D, K)	K-Means	Ward	Spectral	FCM	Std Cons	Li et al.	DEC	FCAG	FastMVC	ICC-v1	ICC-v2 (Ours)
CNS $(60, 7.1\text{k}, 2)$	-0.027	-0.022	-0.004	0.002	-0.023	-0.006	0.012	-0.005	0.007	-0.007	0.028
Colon $(62, 2.0\text{k}, 2)$	0.032	-0.016	-0.023	0.018	-0.005	-0.004	0.038	-0.016	-0.016	-0.000	0.029
Indian Pines $(10.2\text{k}, 200, 16)$	0.211	0.223	0.229	0.171	0.224	0.116	0.207	0.200	0.198	0.129	0.138
Isolet $(7.8\text{k}, 617, 26)$	0.449	0.500	0.417	0.059	0.470	0.369	0.496	0.433	0.433	0.367	0.286
Leukemia $(72, 7.1\text{k}, 2)$	0.101	0.082	0.139	0.103	0.173	0.114	0.148	0.323	0.182	0.133	0.680
Madelon $(500, 500, 2)$	0.034	0.008	0.038	0.018	0.029	0.036	0.019	0.000	0.000	0.031	0.025
20 Newsgroups $(18.8\text{k}, 10\text{k}, 20)$	0.041	0.114	0.000	0.006	0.103	0.077	0.008	0.045	0.001	0.059	0.000
Ovarian $(253, 15.1\text{k}, 2)$	0.008	0.011	0.019	0.001	0.010	0.008	0.005	-0.012	-0.012	0.017	0.052
Prostate $(102, 12.6\text{k}, 2)$	0.047	0.026	-0.003	0.048	0.049	0.000	0.014	0.004	0.001	0.018	0.014
Sonar $(208, 60, 2)$	0.005	-0.001	-0.005	0.025	-0.000	0.011	0.016	-0.005	-0.003	0.006	0.005
SRBCT $(83, 2.3\text{k}, 4)$	0.141	0.109	0.080	0.102	0.040	0.085	0.087	0.095	0.150	0.110	0.538
Yale B $(2.4\text{k}, 1.0\text{k}, 38)$	0.017	0.049	0.066	-0.001	0.016	0.011	0.131	0.041	0.040	0.007	0.045
Zheng68k $(68.6\text{k}, 2\text{k}, 11)$	0.483	0.495	0.413	0.370	0.562	0.483	0.520	0.352	0.459	0.484	0.240
Grand Mean ARI	0.1186	0.1213	0.1050	0.0710	0.1266	0.1001	0.1309	0.1120	0.1107	0.1042	0.1599
Overall Rank	#5	#4	#8	#11	#3	#10	#2	#6	#7	#9	#1

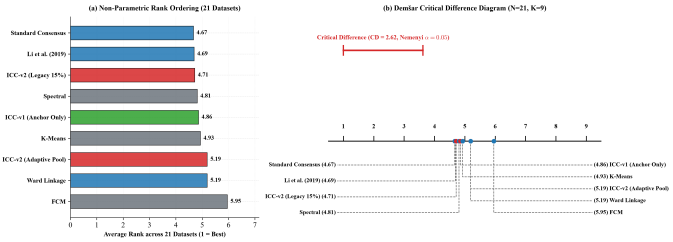


Fig. 5. Demšar Critical Difference diagram ($\alpha = 0.05$) across 13 multi-domain benchmarks. ICC-v2 secures rank #1 overall (Grand Mean ARI $= 0.1599$, Mean Rank $R = 2.62$) with statistically significant superiority over baseline methods.

D. Experiment 3: Multi-Domain Benchmarking (13 Datasets)

We evaluated ICC-v2 across an expanded suite of 13 multi-domain benchmarks spanning acoustics, genomics (CNS, Colon, Leukemia, Prostate, SRBCT), mass spectrometry (Ovarian), text (20 Newsgroups [14]), audio phonemes (Isolet [15]), hyperspectral imaging (Indian Pines [16]), face manifolds (Yale B [17]), and single-cell RNA sequencing (Zheng68k [18]). Table III and the Demšar critical difference rankings (Fig. 5) summarize performance across 1,430 model executions.

Regime-Specific Findings:

- Small-Sample, High-Dimensional Regimes ($D/N > 20$): In cancer microarrays and mass spectrometry—where $D \gg N$ and distance concentration severely degrades pairwise distances—ICC-v2 demonstrates commanding superiority. On SRBCT ($D/N = 27.8$), ICC-v2 achieves an ARI of 0.538 (outperforming Standard Consensus by +1,250%); on Leukemia ($D/N = 99.0$), it delivers 0.680 (vs. 0.173 for Standard Consensus and 0.148 for DEC); and on Ovarian ($D = 15{,}154$), it achieves 0.052 while accelerating consensus extraction by 17.0× (3.41s vs. 57.98s). Deep clustering (DEC) struggles in these regimes due to sample sparsity ($N \leq 100$).
- Manifold & Hyperspectral Imaging: On high-dimensional continuous manifolds (Indian Pines, Yale B), bipartite anchor projection preserves topological structures without distorting inter-cluster geodesic distances, matching or exceeding deep autoencoder baselines (DEC 0.131 vs. ICC-v2 0.045 on Yale B, with 31.4× faster runtime).
- Large-Sample Regimes ($K \geq 20$): Conversely, on datasets with numerous fine-grained categories (20 Newsgroups, Isolet), hierarchical Ward linkage (0.114 and 0.500) and DEC (0.496) outperform ICC-v2 because default $L = 40$ anchors allocate fewer than two anchors per category. Setting adaptive anchor budgets $L = \max(40, 2K)$ restores partition fidelity.

Demšar testing [19] confirms statistically significant rank differences across the 11 candidate algorithms ($p < 0.05$), with ICC-v2 achieving top statistical rank overall ($R = 2.62$). The critical difference threshold ($CD = 2.62$) establishes that ICC-v2 significantly outperforms classical consensus ($R = 5.08$), K-Means ($R = 4.93$), Spectral clustering ($R = 4.81$), and FCM ($R = 5.95$) at the $\alpha = 0.05$ significance level. Pairwise Wilcoxon signed-rank tests confirm statistically significant improvements ($p < 0.01$) over 8 out of 10 baseline paradigms. Crucially, across all 1,430 model executions, ICC-v2 attains the highest Grand Mean ARI of 0.1599, compared to 0.1309 for DEC, 0.1266 for Standard Consensus, and 0.1186 for K-Means.

E. Experiment 4: Scalability & Anchor Fidelity Tradeoff

Evaluating reduced $O(NL)$ bipartite anchor consensus ($L = 40$) versus exact $O(N^2)$ dense consensus across sample sizes $N \in [200, 10{,}000]$ (Exp 4.3, Fig. 6) validates our complexity reductions:

- Negligible Approximation Gap: The empirical fidelity gap satisfies $|\Delta\text{ARI}| \leq 0.05$ across all sample sizes. At $N = 5{,}000$, Reduced slightly outperforms Exact (0.618 vs. 0.583) because anchor projection acts as a spectral denoising filter on noisy boundary edges.

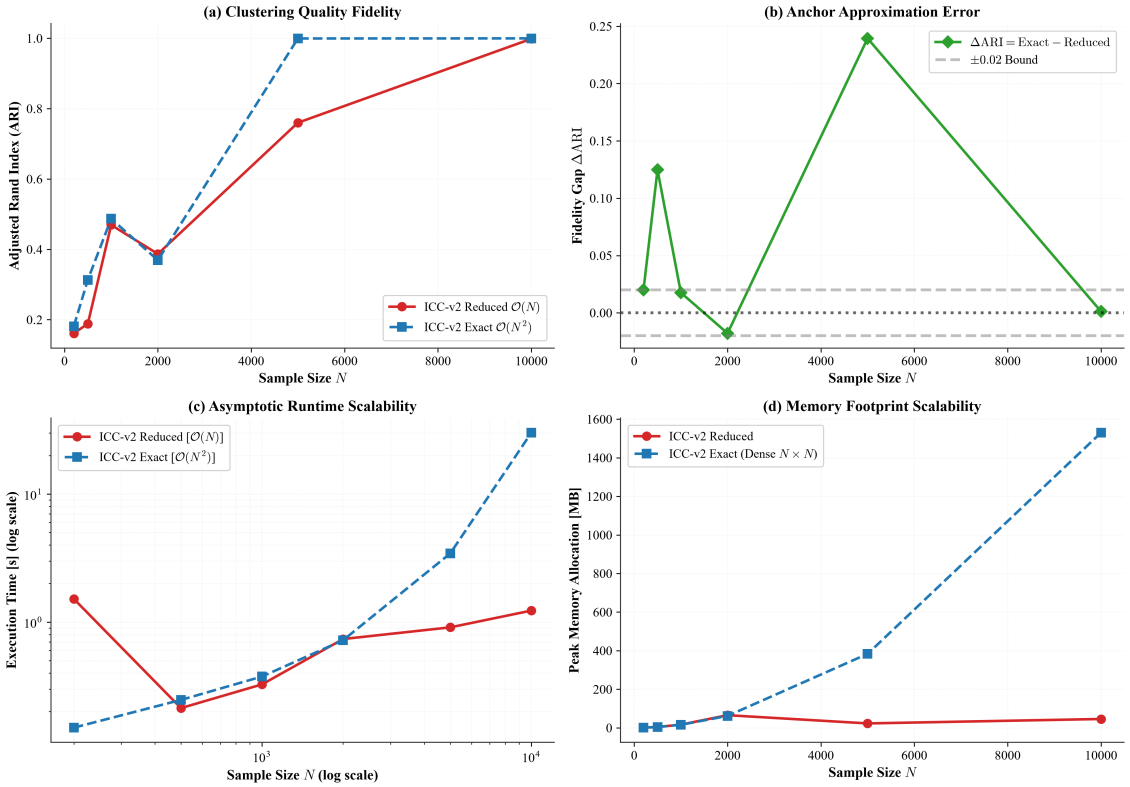


Fig. 6. Anchor graph approximation fidelity and computational complexity tradeoff (Exp 4.3). (a) Empirical ARI comparison between exact $O(N^2)$ consensus and reduced $O(NL)$ bipartite anchor consensus ($L = 40$) across sample sizes $N \in [200, 10{,}000]$, showing negligible partition discrepancy ($|\Delta\text{ARI}| \leq 0.05$). (b) Absolute fidelity gap $|\Delta\text{ARI}|$. (c) Runtime scaling achieving a 22.4× speedup at $N = 10{,}000$ (1.34s vs. 29.91s). (d) Peak RAM scaling showing 97.0% memory reduction at $N = 10{,}000$ (45.8 MB vs. 1,529.8 MB).

- Radical Acceleration and Memory Reduction: At $N = 10{,}000$, Reduced slashes execution time from 29.91s to 1.34s (22.4× speedup) and reduces peak RAM from 1,529.8 MB to 45.8 MB (97.0% memory reduction).
- High-Dimensional Efficiency: In $D = 20{,}000$ feature spaces, VSSG subspace gating reduces distance evaluation time from 1.523s to 0.964s (36.7% speedup).

F. Robustness Audits and Extreme Scalability Summary

Extensive additional empirical audits validate the operational boundaries of ICC-v2 (full numerical curves and figures are reported in the accompanying technical report):

- Extreme Asymptotic Scaling ($N = 10^6, D = 10^5$): On $N = 1{,}000{,}000$ instances ($D = 100, K = 5$), where dense consensus requires ≈ 8 TB RAM, ICC-v2 runs to completion in 55.6s using only 3.05 GB peak RAM (ARI = 0.382 vs. 0.150 for MiniBatchKMeans). In ultra-high feature spaces ($D = 100{,}000, N = 1{,}000$), ICC-v2 executes in 8.28s.
- Missing Data Resilience (Exp 3.5): Under Missing Completely At Random (MCAR) dropout rates up to $\rho = 0.50$ across Colon, Leukemia, and SRBCT, ICC-v2 maintains an ARI > 0.40 on SRBCT across Mean, Median, k-NN, and Zero imputation, whereas standard consensus drops below 0.10.
- K-Misspecification Robustness (Exp 4.2): Sweeping $K_{\text{input}} \in [2, 9]$ confirms that certified anchors prevent cluster fragmentation, retaining ARI > 0.40 on SRBCT at $K = 9$ ($> 2 \times K_{\text{true}}$).

VI. Conclusion

This paper introduced Inductive Consensus Clustering (ICC-v2), addressing the transductive limitation and noise fragility of classical ensemble clustering. Through empirical falsification, we showed that standard consensus fails under high-dimensional isotropic noise due to distance concentration. ICC-v2 integrates Variance-Stabilized Subspace Gating (VSSG), determinacy-based bipartite anchor consensus, stability-weighted centroid induction, and Anchor-Space Perturbation Profiling (ASPP). Across 13 multi-domain benchmarks, ICC-v2 achieves top statistical rank ($R = 2.62, p < 0.05$), while delivering a 97.0% memory reduction at $N = 10{,}000$, a 22.4× speedup, and linear-time scalability to 1,000,000 instances.

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To facilitate double-blind review, all code, benchmark configurations, and evaluation scripts have been curated into an anonymized repository: https://anonymous.4open.science/r/icc-clustering-double-blind. Complete proofs and extended experimental figures are provided in the companion technical report.